

Advanced (Habanero)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** If $y = 3x - 2$ and $y = mx + b$ are perpendicular, what is m ?
(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 3
(D) -3
(E) 0
- Q3.** Which equation represents a line perpendicular to $y = -4x + 2$?
(A) $y = \frac{1}{4}x + 1$
(B) $y = -\frac{1}{4}x - 2$
(C) $y = 4x - 3$
(D) $y = -4x + 5$
(E) $y = \frac{1}{4}x - 2$
- Q4.** What is the equation of a line that is perpendicular to $y = x + 2$ and passes through $(1, 3)$?
(A) $y = -x + 4$
(B) $y = x + 1$
(C) $y = -x + 2$
(D) $y = x - 2$
(E) $y = -x - 1$
- Q5.** If two lines have slopes m_1 and m_2 , what condition must be met for them to be perpendicular?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 + m_2 = 0$
(E) $m_1 - m_2 = 0$
- Q6.** What is the equation of the line that is perpendicular to $y = 2x - 1$ and has a y -intercept of 3?
(A) $y = -\frac{1}{2}x + 3$
(B) $y = \frac{1}{2}x + 1$
(C) $y = 2x - 2$
(D) $y = -2x + 4$
(E) $y = -\frac{1}{2}x - 1$
- Q7.** If the equation $y = mx + b$ is perpendicular to the line $y = 2x + 1$, what is the value of m ?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q8.** Which of the following lines is perpendicular to $y = -3x + 2$?
(A) $y = \frac{1}{3}x - 1$
(B) $y = -\frac{1}{3}x + 2$
(C) $y = 3x - 2$
(D) $y = -3x + 1$
(E) $y = \frac{1}{3}x + 1$
- Q9.** Find the equation of a line perpendicular to $y = x - 2$ that passes through the point $(2, 5)$. Show all work.
- Q10.** Given that $y = 2x + 1$ and $y = mx + b$ are perpendicular lines, and the point $(1, 3)$ lies on the line $y = mx + b$, find the values of m and b . Show all work.

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line perpendicular to $y = x - 2$ that passes through the point $(2, 5)$. Show all work.
- Q10.** Given that $y = 2x + 1$ and $y = mx + b$ are perpendicular lines, and the point $(1, 3)$ lies on the line $y = mx + b$, find the values of m and b . Show all work.

Chef's Tips

- Provide clear instructions to students.
- Ensure students have necessary materials.
- Allow students to ask questions before starting the test.